

Quantum-Steered Cognition

An Auditable Quantum-Entropy Channel for Autonomous AI Agents - and the Experiment It Makes Possible

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Executive Summary

For a century, experiments have asked whether mind can bias a random number generator. The field is deadlocked in a peculiar way: its cumulative statistics are striking - the Global Consciousness Project's formal series closed at $Z = 7.31$ over 500 pre-registered events - yet every major positive claim has dissolved under one of three human failure modes: someone chose which events counted, someone chose the analysis after seeing data, or an artifact imitated the effect.

Ability AI operates Cornelius, a production knowledge-work agent whose exploratory cognition is steered by live quantum randomness: vacuum-fluctuation measurements from the ANU quantum random number generator, mapped through a fixed 192-byte protocol into directions in the agent's knowledge graph. The channel enforces five invariants: quantum-only bytes with abort on unavailability, context declared before measurement, no client-side conditioning, complete append-only draw provenance, and segregated non-endorsed output.

This architecture is, incidentally but exactly, the instrument the deadlocked literature lacked: an experimenter that cannot select, cannot forget, and can be cloned with only its randomness swapped. We propose a pre-registered two-arm experiment - quantum-steered sessions against pseudorandom-steered sessions in an identical pipeline, with blinded rating, fixed N , and Bayesian analysis. The expected result is null: physics says quantum and pseudorandom bytes are functionally indistinguishable, and the null itself certifies the channel as a bias-free serendipity engine.

The recommendation stands regardless of how the question resolves. The design pattern - auditable entropy, an empty file-drawer by construction, provenance-complete logs - pays for itself in any AI system whose stochastic choices may later need defending. We claim no experimental result: the four logged sessions demonstrate that the pipeline runs, and nothing else.

The Problem and Why Now

A century of striking statistics that never survive their audits

The random number generator is the oldest quantitative instrument for the mind-matter question: if intention or attention can bias a physically random process, a device with a precisely known output distribution becomes a detector (Jahn, 1982). The resulting record is extreme in both directions. On one side: a meta-analytic database of 380 intention experiments (Bösch et al., 2006) and a 17-year global RNG network whose formal event series closed at $Z = 7.31$ over 500

events - odds of roughly 3 trillion to one against chance (Nelson, 2024). On the other: the same meta-analysis found the pooled effect's sign flips between statistical models and its magnitude shrinks as studies get bigger across four orders of magnitude, the classic signature of publication bias, ending in a verdict of "not proven" (Bösch et al., 2006); the global network's own senior analyst concluded after 17 years that "the data do not support the global consciousness proposal" (Bancel, 2017); and the most publicized direct observer-effect experiment was refuted when sham sessions - same apparatus, no observer - reproduced the effect (Walleczek & von Stillfried, 2019).

Read as a design specification rather than a verdict, the literature's failures reduce to three human degrees of freedom:

1. **Selection freedom.** A motivated human decides which events enter the formal record. In the Global Consciousness Project's registry, one person contributed 234 of 300 formal events at a combined $Z = 5.91$, while everyone else's 66 events reached only $Z = 1.26$ (May & Spottiswoode, 2011).
2. **Analysis freedom.** Endpoints and models chosen after seeing data. This is the mechanism behind the famous precognition series - nine experiments, over 1,000 participants, published in a flagship journal (Bem, 2011) - that failed in every independent pre-registered replication (Galak et al., 2012; Muhmenthaler et al., 2024; Ritchie et al., 2012) and whose evidence a Bayesian re-analysis rated "weak to nonexistent" (Wagenmakers et al., 2011).
3. **Channel opacity.** Hardware and post-processing artifacts that imitate the sought effect, as in the sham-session refutation above (Walleczek & von Stillfried, 2019) and the calibration arithmetic showing that small hardware bias inflates the target statistic unless calibration runs are far longer than practitioners use (Steiglitz, 2023).

The second problem: agents explore where their training points

Autonomous AI agents have a structurally unrelated problem with the same shape. An agent exploring a large knowledge base goes where its learned retrieval weights, embeddings, and habits point; every path is shaped by algorithmic and motivational bias. Randomness is a standard, rational tool for exploration in computation and cognition (Gershman, 2021) - but pseudorandomness inherits the auditability problem: a seed and an algorithm chosen by someone, somewhere, with no physical guarantee behind the draw.

Cornelius's entropy channel was built for this second problem: uniform, algorithm-unbiased exploration of a 5,491-note knowledge graph. It was designed, however, under an explicit speculative hypothesis held by its author: *if anything can influence quantum measurement outcomes, a bias-transparent channel from quantum bits to cognitive direction is a door such influence could walk through*. Under the null hypothesis the channel is simply a clean serendipity engine. Both readings make the same architecture valuable.

Why now

Two 2026 events frame the moment. The original Global Consciousness Project's data collection ended on 3 April 2026 after a hardware failure, with a successor citizen-science network (GCP 2.0) planned at 1,000 devices hosting 4,000 RNGs (*Global Consciousness Project 2.0*, n.d.). And the first fully pre-registered, selection-free test of the GCP paradigm - algorithmic event selection, insert-only ledger, registered 2026-07-03, $n = 180$ scored days - is now running (Holmberg, 2026). The

field's own best move is to remove the human from the loop. An autonomous agent is the natural next substrate for that move.

The Approach: An Auditable Quantum-Entropy Channel

The system in one paragraph

Cornelius is a production knowledge-management agent (a “second brain” over an Obsidian knowledge graph: 5,491 indexed notes, 49,551 edges at the July 2026 index build) whose cognitive schedules run on the claude-fable-5 language model. On a fixed schedule it runs exploration sessions steered by true quantum randomness: a live measurement of electromagnetic vacuum fluctuations at the Australian National University - quantum noise clearing electronic noise by a constant 8.5 dB (Symul et al., 2011) - is delivered over the public QRNG API (*API Documentation - ANU QRNG*, n.d.) and mapped, without any client-side hashing or whitening, into two knowledge-graph neighborhoods, one analytical move, and one interpretive lens. Figure 1 shows the full path from vacuum to cognition.

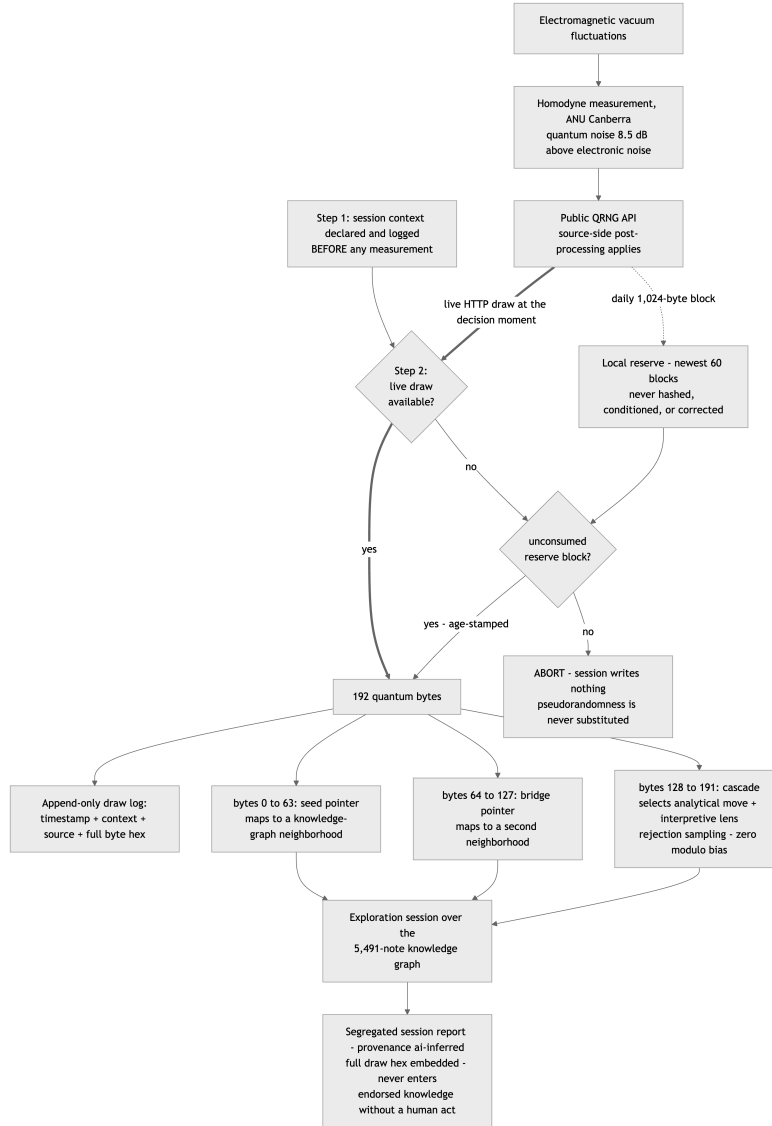


Figure 1. The entropy channel, from vacuum fluctuations to a segregated session report. The heavy path is the preferred live draw; the reserve is an outage fallback; if no quantum bytes are available the session aborts. Rendered 2026-07-14 from the system snapshot of 2026-07-13.

The mechanics, briefly. A daily job fetches one 1,024-byte block into a local reserve (newest 60 blocks) and stamps channel health; it is forbidden to hash, fold, or correct the bytes - observed block-mean outliers are recorded, never fixed. A session begins by composing its context - the date, the open question - as a string, *before* any measurement fires. The draw script then attempts a live draw; on failure it falls back to the newest unconsumed reserve block; if neither source is available it aborts and writes nothing. Each session consumes 192 bytes, sliced deterministically: bytes 0 to 63 map through monotone transforms into a *seed* neighborhood of the knowledge graph, bytes 64 to 127 into a second *bridge* neighborhood, and bytes 128 to 191 select the analytical move (by rejection sampling, eliminating modulo bias) and the interpretive lens (a 16-way modulo over

256 values - exactly zero bias). Every draw is logged append-only with its timestamp, declared context, source, and complete byte string in hex.

The five invariants

1. **Quantum-only.** No quantum bytes, no session. Pseudorandomness is never silently substituted; one substitution would corrupt the interpretability of the entire draw record.
2. **Context before measurement.** The session’s question is fixed and logged before the physical measurement occurs, giving every draw a declared referent.
3. **No client-side conditioning.** Every mapping from bytes to choices is monotone or exactly bias-free. A weak statistical tilt in the bytes would surface as a coherent semantic drift rather than being scrambled away.
4. **Full provenance.** Every draw is logged before any outcome exists. The file-drawer is empty by construction.
5. **Segregated output.** Session reports carry provenance *ai-inferred*, embed the full draw hex, and never enter the agent’s endorsed knowledge without a separate human act.

Where the transparency claim honestly ends

Invariant 3 covers the client side only. The served ANU stream is post-processed at the source: at minimum by documented distributional shaping (Symul et al., 2011), and per a later independent description, by cryptographic hashing (Kavulich et al., 2021). Those two descriptions are not equivalent, and no authoritative current statement says which applies to the public API’s bytes (*API Documentation - ANU QRNG*, n.d.). The source is also *trusted*, not *certified*: its randomness guarantee is a physics argument about the device, not a Bell-test bound of the kind modern quantum information science can provide (Liu et al., 2018; Pironio et al., 2010). Both facts are carried into the experimental design below rather than papered over: hypotheses are pitched at the level of *which bytes arrive*, not waveform structure, and a certified-randomness upgrade is the pre-declared final rung of the escalation ladder.

Evidence: What Is Actually Established

This section states findings before arguing what they mean. Table 1 summarizes the six evidence streams the question touches, with each stream’s strongest positive claim and its strongest countervailing evidence side by side.

Table 1. The six evidence streams: headline positive claim, strongest countervailing evidence, and status. Full quote-level support in the research bundle’s evidence notes.

Evidence stream	Headline positive claim	Strongest countervailing evidence	Status
Intention-RNG experiments (1959 to 2006)	Cumulative database deviates from chance across hundreds of studies (D. I. Radin & Nelson, 1989; D. Radin & Nelson, 2000)	Meta-analytic effect is tiny, extremely heterogeneous, and consistent with small-study/publication bias; verdict “not proven” (Bösch et al., 2006); pre-registered consensus-protocol replications null (Walach et al., 2022)	Contested; unreplicated on demand

Evidence stream	Headline positive claim	Strongest countervailing evidence	Status
Global Consciousness Project (1998 to 2026)	Formal event series reaches $Z = 7.31$ over 500 events (Nelson, 2024)	Arithmetic undisputed; structure favors goal-oriented experimenter selection over a passive field (Bancel, 2017; May & Spottiswoode, 2011); XOR architecture argument (Shrikumar, 2023)	Deviation real, cause contested
Consciousness-collapse theory	Consistent collapse models can be formulated and are testable (Chalmers & McQueen, 2022)	Consciousness unnecessary for measurement (Yu & Nikolić, 2011); brain-scale coherence decoherence-forbidden (McKemmish et al., 2009; Tegmark, 2000)	Speculative; constrained, not excluded
Direct observer-effect experiments	Double-slit interference shifts with attention (D. Radin et al., 2012)	Sham sessions without observers reproduce the effect; false-positive mechanism identified (Walleczek & von Stillfried, 2019); interpretation conflicts with the projection postulate (de Bianchi, 2013)	Unsupported after re-analysis
Quantum vs pseudo randomness in use	Quantum sources offer in-principle unpredictability (Herrero-Collantes & Garcia-Escartin, 2017)	No statistical or algorithmic test distinguishes ANU QRNG output from good PRNGs at 168 Gbit per generator (Kavulich et al., 2021); no functional difference in application tests (Mogos, 2016)	Functional equivalence is the baseline
Anomalous-cognition experiments (rigor benchmark)	Nine-experiment series with about 1,000 subjects, mean $d = 0.22$ (Bem, 2011)	Independent replications null (two of them pre-registered) (Galak et al., 2012; Muhmenthaler et al., 2024; Ritchie et al., 2012); Bayesian re-analysis finds evidence “weak to nonexistent” (Wagenmakers et al., 2011)	Canonical false-positive case study

Three facts in this table do the heaviest work for what follows.

The statistics are real; the causes are contested. No published critic disputes the Global Consciousness Project’s arithmetic. The dispute is entirely about the source of the structure: a passive global field, a goal-oriented experimenter effect concentrated in whoever selects the events (Bancel, 2017; May & Spottiswoode, 2011), or accumulated small biases. The honest reading is a preserved contradiction, not a verdict - and the proponents have a substantive rejoinder of their own on the meta-analytic side: if the true effect were a constant z -shift rather than a constant per-bit force, the inverse-scaling with sample size would be expected rather than diagnostic of bias (Bösch et al., 2006). The deadlock is structural. Only designs that remove the contested human freedoms can break it.

The physics baseline is functional equivalence. Across 100 strings of 1.68×10^9 bits per generator, no statistical or algorithmic-randomness test distinguished ANU quantum output from four good pseudorandom generators - while the same battery did detect deliberately introduced structure, so the null is not test blindness (Kavulich et al., 2021). An application-level comparison likewise found no functional difference (Mogos, 2016). Under standard physics, swapping the entropy source changes nothing measurable downstream. Any reliable behavioral difference between a quantum-steered and a pseudorandom-steered agent would therefore be anomalous by construction - which is exactly what makes the experiment falsification-ready.

The channel runs. As of the channel-health stamp of 2026-07-13: status `api_ok`; 4 reserve blocks held, 3 of them unconsumed; last block mean 128.67 against the 127.5 reference expectation (recorded, never corrected - see invariant 3); 0 consecutive failures; 6 draws in the append-only log; 4 completed session reports with full draw hex embedded. Consistent with this whitepaper’s protocol-only scope, these sessions demonstrate end-to-end feasibility and support no inference about outcomes - no outcome claim appears anywhere in this document.

The Proposed Experiment: A Pre-Registered Two-Arm Test

The question in its cleanest form: does an autonomous agent consuming quantum bytes behave measurably differently from an identical agent consuming pseudorandom bytes, when physics says it must not (Kavulich et al., 2021)? Table 2 is the design’s core argument - every feature answers a named, documented failure of the prior literature; Table 3 fixes the parameters, and Figure 2 shows the whole design end to end.

Table 2. Documented failure modes of prior mind-matter experiments and the design feature that removes each.

Documented failure mode	Source of the diagnosis	Design feature that removes it
Post hoc event selection by a motivated experimenter	GCP formal-event asymmetry: selector’s 234 events $Z = 5.91$ vs others’ 66 events $Z = 1.26$ (Bancel, 2017; May & Spottiswoode, 2011)	Sessions fire on a fixed schedule; arm assignment from a pre-committed public randomization list; no human chooses when or what to measure
Analysis flexibility / exploratory results presented as confirmatory	Bem case and its Bayesian re-analysis (Bem, 2011; Wagenmakers et al., 2011)	Analysis code, endpoints, and priors frozen at pre-registration; exploratory analyses labeled and Bonferroni-style discounted (Muhmenthaler et al., 2024)
Optional stopping and data peeking	Meta-analytic small-study signature (Bösch et al., 2006)	Fixed session count declared in advance; Bayes factors reported at the pre-declared N only
Calibration and bias-inflation artifacts in physical-RNG tests	Bias-inflation mechanism and calibration-length requirement (Steiglitz, 2023); process-detail dependence (Ibison, 1998)	Control arm consumes bytes through the identical byte protocol and pipeline; byte-level statistics of both arms monitored against reference expectation
Environmental or hardware artifacts masquerading as anomalies	Sham-session false positive in the double-slit series (Walleczek & von Stillfried, 2019)	Sham sessions (pipeline run with no cognitive consumer) included as a third condition in the audit set;

Documented failure mode	Source of the diagnosis	Design feature that removes it
		hardware health continuously stamped
Unblinded evaluation of outcomes	General; observer-expectancy family (Walleczek & von Stillfried, 2019)	Session reports stripped of arm labels before rating; human plus LLM-panel evaluators blind to source
File-drawer selective reporting	1,544-study filedrawer simulation (Bösch et al., 2006)	Every session is logged append-only before its content exists; the ledger is insert-only and complete by construction (Holmberg, 2026)
Silent substitution or corruption of the entropy source	Channel-integrity requirement (this system)	PRNG-abort invariant on the production channel; full draw provenance (timestamp, declared context, source, complete byte hex) makes every session re-derivable

Table 3. Protocol parameters fixed at pre-registration.

Parameter	Specification
Design	Two-arm, blinded-evaluation, fixed-N comparison of exploration sessions
Arm Q (experimental)	Production channel: live ANU QRNG draw, 192 bytes, context declared before measurement; reserve fallback source-stamped
Arm P (control)	Protocol-scoped sandbox: seeded CSPRNG bytes through the identical byte protocol, pointer mapping, cascade, and report format; production draw log and reserve untouched
Audit condition	Sham sessions: full pipeline run with no cognitive consumer, both byte sources
Unit of analysis	One exploration session (one 192-byte draw, two neighborhoods, one move-lens pair, one report)
Primary endpoint	Pre-defined composite of blinded ratings: usefulness, novelty, groundedness of session findings
Secondary endpoints	Byte-level statistics of Arm Q draws vs reference expectation (mean, runs, autocorrelation); context-congruence of landed neighborhoods
Assignment	Pre-committed public randomization list, generated and deposited at registration
Blinding	Arm labels stripped from reports before evaluation; evaluators (human and LLM panel) blind to source
Sample size	Fixed N declared at registration; sized for BF10 of at least 10 or BF01 of at least 10 sensitivity under the registered priors (a formal design analysis is part of registration)
Analysis	Default Bayes factors plus posterior estimation with pre-declared priors; no optional stopping; exploratory analyses labeled
Positive-result path	No anomaly claim from one system: artifact audit, then independent replication, then certified-randomness upgrade of the source (Liu et al., 2018; Pironio et al., 2010)
Registration	Time-stamped public deposit of protocol, code, priors, and randomization list before the first session (Holmberg, 2026)

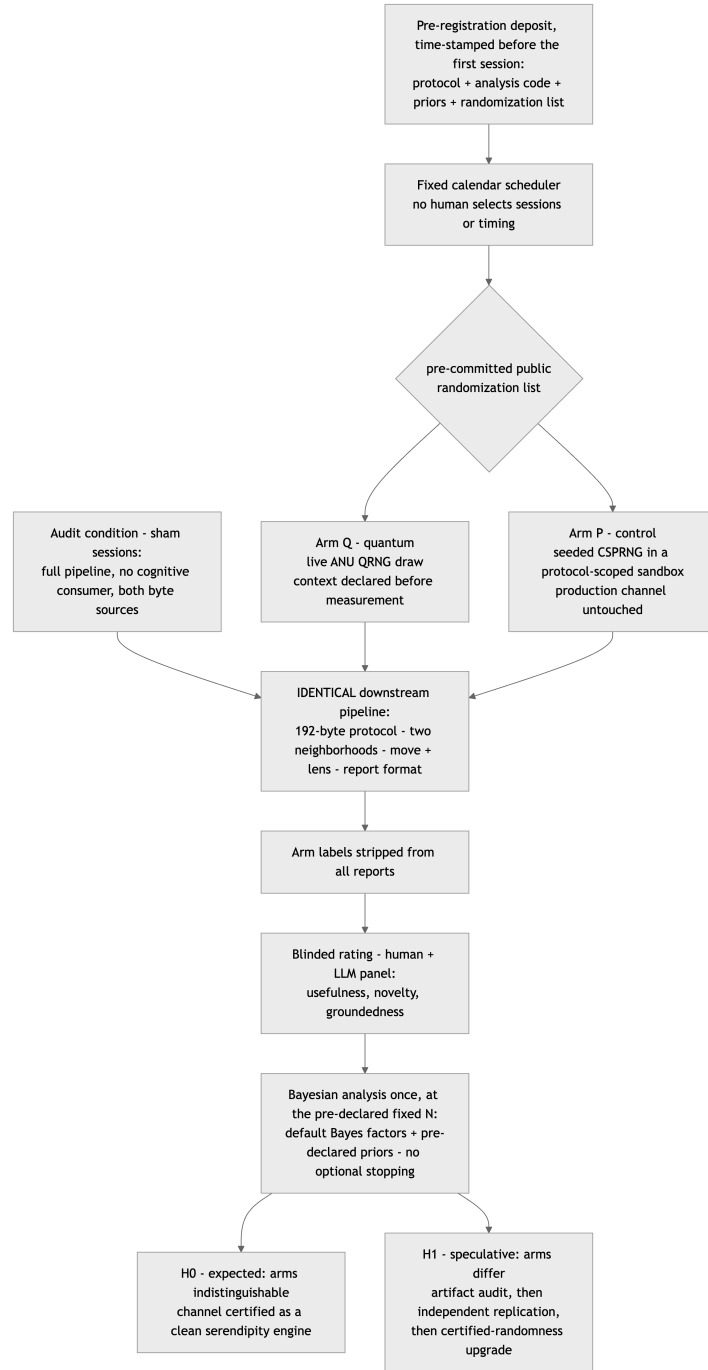


Figure 2. The two-arm protocol. Both arms and the sham audit condition feed an identical downstream pipeline; evaluation is blinded; analysis happens once, at the pre-declared N.

The two hypotheses, priced honestly. H0, the expected outcome: session outcomes are statistically indistinguishable between arms - the firm prediction of standard physics (Kavulich et al., 2021; Mogos, 2016) and of the computational theory of randomness use (Gershman, 2021). H1, the speculative alternative: quantum-steered sessions differ on the pre-declared endpoints. H1 is

motivated by the information-field conjecture in the system’s design, not by the prior literature, which on our reading does not establish the effect (Bancel, 2017; Bösch et al., 2006; Walach et al., 2022). One inversion is worth an executive’s attention: a positive result would *not* vindicate the popular “consciousness collapses the wavefunction” story - collapse models preserve the standard quantum statistics, so a statistical deviation would refute rather than confirm them (de Bianchi, 2013). H1 is a claim about outcome statistics, agnostic as to mechanism.

A positive result licenses no anomaly claim from this system alone. The pre-registered escalation path is: artifact audit against the sham-session record and byte-level statistics; independent replication on other agent instances; and only then migration toward certified randomness, where the min-entropy guarantee comes from a Bell-inequality violation rather than device trust (Bierhorst et al., 2018; Liu et al., 2018; Pironio et al., 2010) - accepting a rate cost of roughly seven orders of magnitude.

Implications and Recommendations

For AI builders: the design pattern pays for itself under the null

Every recommendation here traces to the Evidence section, and none depends on the exotic hypothesis being true.

1. **Adopt auditable-entropy invariants for any stochastic component you may later need to defend.** The file-drawer problem that consumed a century of RNG research (Bösch et al., 2006) is not a parapsychology problem; it is what happens to any system whose random choices are logged after outcomes are known. Declare context before the draw, log append-only with full provenance, never condition the stream silently, abort rather than substitute. These four habits cost almost nothing at design time and are unbuildable retroactively.
2. **Treat the null result as product value, not failure.** The physics baseline says the arms will not differ (Kavulich et al., 2021; Mogos, 2016). Under that expected outcome, the experiment certifies the channel as a clean serendipity engine: uniform exploration of the knowledge graph, free of the algorithmic and motivational biases that shape every learned retrieval path (Gershman, 2021). To our knowledge it would also be the first pre-registered demonstration that an autonomous agent’s outputs are invariant to entropy source at behavioral resolution.
3. **Pre-register before claiming.** The entire post-2011 reformation - registration before data, fixed N, blinded evaluation, Bayesian model comparison (Wagenmakers et al., 2011) - is the entry fee for any claim in this space. An agent that runs its own pre-registered self-experiment on a scheduler, at marginal cost near zero, makes that fee trivial to pay. Never claim from one system: replication is a pre-declared rung, not an afterthought.

For researchers: structural rather than forensic confound removal

Where the Global Consciousness Project debate must reconstruct selection effects forensically from a 17-year registry (Bancel, 2017; May & Spottiswoode, 2011), this design precludes them structurally: the scheduler selects, the ledger is complete before content exists, the evaluators are blind. The design is deliberately convergent with the field’s own best current move - the pre-registered, selection-free GCP 2.0 test now running (Holmberg, 2026) - but moves the substrate

from network statistics against world events to a *cognitive consumer* of quantum bytes against its own pseudorandom twin. The two designs fail differently and are therefore mutually informative.

One candid limit, preserved rather than hidden: if the operative hypothesis in the historical data is *experimenter* goal-orientation in the Decision Augmentation sense (May & Spottiswoode, 2011), automation does not obviously remove goal-orientation - it relocates it, to the system's designer and perhaps to the agent itself, whose declared contexts are goal statements. The design answers this by making goal-orientation *measurable* (the context-congruence endpoint) rather than by claiming to have eliminated it. A positive result would establish an anomaly and sharpen, not settle, the source question (Bancel, 2017).

Limitations

This whitepaper describes a protocol, not results; the 4 logged sessions demonstrate pipeline feasibility only, and no outcome data exist. The entropy source is trusted, not certified: its quantum dominance (8.5 dB clearance) is a physics argument about the device (Symul et al., 2011), the served stream carries source-side post-processing whose exact nature is unresolved (*API Documentation - ANU QRNG*, n.d.; Kavulich et al., 2021), and hypotheses about physical-layer structure are out of reach of the public stream regardless. The effect-size ceiling from the prior literature is brutal - any real per-bit effect sits in the fifth decimal place (Bösch et al., 2006) - so the experiment is powered for behavioral-scale differences only; a null leaves microscopic effects unaddressed. The evaluator panel includes LLM raters whose biases are imperfectly characterized. And the conflict of interest is total and disclosed: the system has one designer, whose intentions saturate its architecture, who is the author of this whitepaper and its sponsor; the drafting agent is the system described. The goal-orientation critique applies to us with full force and is answerable only by independent replication.

Conclusion

A production AI agent, built to explore a knowledge graph without algorithmic bias, turns out to be the instrument a century of mind-matter experiments lacked: an experimenter that cannot select, cannot forget, and can be cloned with only its randomness swapped. The proposed two-arm, blinded, fixed-N, Bayesian protocol converts a deadlock between undisputed statistics and disputed causes into a falsifiable comparison in which every design feature answers a named historical failure. The expected result is null, and the null is useful - it certifies the serendipity engine and closes the agent-cognition instance of the quantum-pseudorandom distinguishability question. The improbable result is specified in advance, priced honestly, and chained to an escalation ladder ending in certified randomness and independent replication. Either way, the question moves from anecdote to logged, pre-registered, repeating measurement - where it should have been all along.

Sponsor, Disclosures, and Availability

Sponsor and conflict of interest. This whitepaper is published by Ability AI. Its author is the designer and operator of the system it describes, and has a professional interest in the architecture's adoption. The dual role is declared above as a limitation, not a footnote.

AI disclosure. This whitepaper was drafted by Claude (model claude-fable-5, Anthropic), operating as the Cornelius agent under the direction of the author, who defined the scope, approved the outline at an explicit gate, and reviewed the final text. The drafting agent is the system described in this whitepaper; this dual role is disclosed as a conflict of interest. Every citation was verified against live bibliographic records (Crossref, OpenAlex, Semantic Scholar); no experimental data exist or were analyzed. Full details in the workspace file `ai-disclosure.md`.

Availability. The entropy-channel implementation, the append-only draw log, the four session reports, and the complete research bundle behind the underlying review (search log, per-source evidence notes with verbatim quotes and locators, evidence matrix, verified bibliography) reside in the author’s agent repository; the channel components are scheduled for public release with the pre-registration deposit. The companion peer-review manuscript, from which this whitepaper is derived, is under journal submission. The ANU QRNG service is publicly accessible (*API Documentation - ANU QRNG*, n.d.).

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